

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari · [Avijeet Singh](#) — Vortx AI · paper doi:10.5281/zenodo.20706893 (v0.1.0; current method: whitepaper v3; emem 2.4.2 at 213e273 and 18adb67) · measurements 29–30 Sep 2026 · scripts: [github.com/Vortx-AI/esa_poster](#)

Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?

emem makes raw satellite observations **addressable by place, band and time**, each a signed record named by the **BLAKE3** hash of its **bytes**. The exact evidence A cites survives compaction, a change of model and a handoff; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name.

fact

bytes

name

cell

time

recall

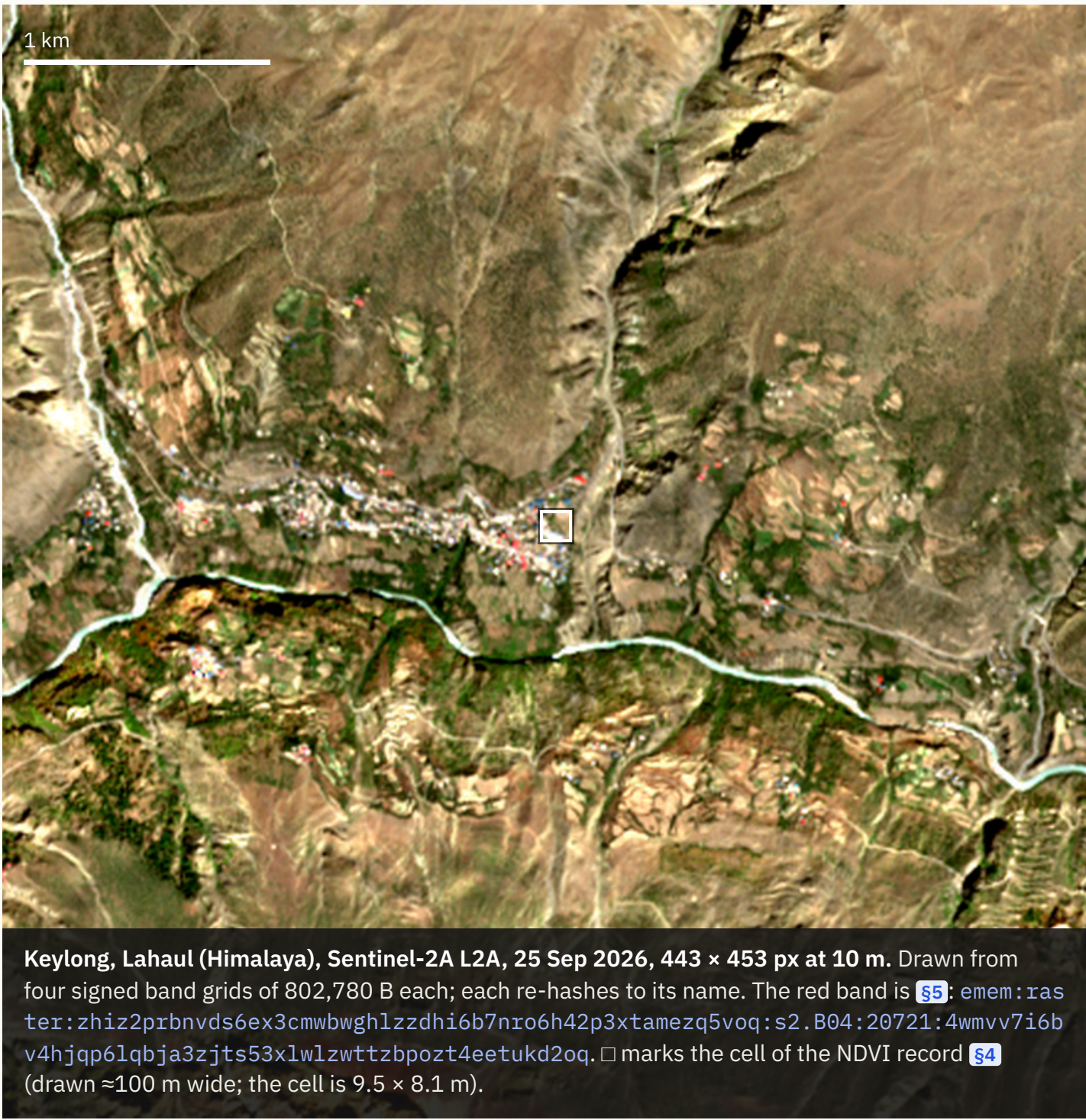
sign

log

receipt

```
f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation{fn_key, args}, privacy, schema_cid, signer, signed_at)
b = CBOR(f), fields in declaration order; NaN → f9 7e00, −0 → +0
fact_cid = base32(BLAKE3-256(b)) · token emem:fact:<cell>:<fact_cid>
lat_q = round((lat+90)/180·(221−1)), lng_q = round((lng+180)/360·(221−1)); pitch 8.583·10−10 = 9.52 × 8.06 m here
tslot = [unix / 86 400] for Sentinel-2 (20721 = 25 Sep 2026)
key k = (cell, band, tslot); as of t: argmaxsigned_at ≤ t (signed_at, cid)
R = Merkle(sorted fact_cids); attestation = Ed25519(PreimageV1("attestation")){R, registry, schema}
leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation))), node = BLAKE3(0x01 || 1 || 1) → RFC 6962 shape
o = Ed25519(BLAKE3("emem.preimage.v10" || request_id || served_at || primitive || cells || fact_cids || merkle))
```

The memory, as coded at emem 18adb67 (fact,rs,cbors,geo.rs,recall,rs,attest,rs,translog,rs,emem-attest). A fact itself carries no signature: the attestation over its batch does, and the log holds the attestation.



2 · FULL TRACE

From one token to the source pixel, without emem software

accept → BLAKE3(b) = cid

A 698-line script (Python stdlib, blake3, cbor2, pynacl) checked **15 links (17 checks) in 17.9 s, all verified**. Grey: what must still be trusted after each check. Ten client paths (REST, MCP, A2A, Python and TypeScript SDKs, LangChain, LlamaIndex, stock blake3) returned the same fact_cid, 3 repetitions each; a tampered mirror copy was rejected.

1–3

bytes → name

4–5

cell binding

6–7

derivation

8–9

upstream

1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... · still trusted: nothing

cell inside the hashed body = cell in the token; cell decodes to the lat/lng the derivation used · nothing

scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order is read from emem's code; no published schema

Planetary Computer STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum; the pixel is the one containing the point (GDAL floor)

emem.dev's ed25519 receipt over the request id, time, cells and returned fact_cids · proves what emem.dev served, not that it is true

entry 2,457,078 is included in the 2,556,451-entry log, whose signed root is consistent with the root geo.qa co-signed · geo.qa is also run by Vortx AI

the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1–5 and 12–14 fit in a 4,906-byte file (our format) that re-verifies 9 checks with networking disabled (unshare -rn) and lists the upstream byte ranges to re-read; a 1-bit tamper, a wrong cell or a wrong key fails it. research/repro/v8/trace_fact.py · verify_bundle.py · [§22](#)

3 · ONE ADDRESS, EVERY PRODUCT

One cell, Keylong [§29](#), holds 207 signed facts from 13 products

emem:cell:defi.zb572.xoso.zbl1ec → current fact per band, each emem:fact:<cell>:<cid> · 13 of 13 re-hash (BLAKE3 of the served CBOR = cid)		
PRODUCT	VALUE	CID
Sentinel-2 B04 reflectance	0.0900	wbbu4ezm...
Sentinel-2 B08 reflectance	0.2502	4utkk3hy...
Sentinel-2 scene class	4 (vegetation)	6kx2hho3...
NDVI (30 Sep scene)	0.4237	3yyaxn5d...
NDWI	-0.5023	juokn77p...
NBR	0.2726	ezs2hnp...
Sentinel-1 RTC VV backscatter	-7.2151 dB	g5sd1fcj...

Read 30 Sep 2026 from GET /v1/cells/defi.zb572.xoso.zbl1ec; 156 of the 207 facts are NDVI. Sources named inside each record: sentinel_s2_l2a, sentinel_s1_rtc_mpc, copernicus.dem.30m.aws, jrc.gsw.v1_4, dmsp_ols.v4.ngdc, met_no. research/repro/data/v11.

5 · HANDOFF: TWO LLMs, EMEM AS THE ONLY MCP SERVER

Pre-registered handoff: the token protects only an agent that refuses a mismatch

B acts → BLAKE3(b) = cid		
A's token		
A's own prose (16 digits)		
prose rounded to "0.47"		
token with a forged cell		

A: Claude Sonnet 5.5 with emem as its only MCP server returned the exact token in 10/10 runs (mean 13 s, 3.3 tool calls; one prose misdated the scene 2025). Rule: irrigate if NDVI ≤ 0.4705, a threshold **set on purpose** 0.0004 below the value so that 2-decimal rounding flips it; it shows the mechanism, not a rate. Primary test, token vs full prose: 10/10 vs 10/10, a tie (p = 1). The rounded arm was added after A's runs (exploratory): Fisher p = 0.0003. **The pre-registered local Qwen2.5-3B as B** (42 of 45 trials run) read 0.4709 from the token yet chose irrigate, 0/19 correct (full prose 10/18); on forged tokens it dropped the cell, resolved the bare id and acted: 0/5 declined.

Pre-registration hashed before trial 1 (research/repro/data/v8/prereg.md). Claude arms: one model family for A and B.

7 · A DRIFT TAXONOMY FROM REAL CASES

Seven ways cited evidence drifted, and the part of the record that shows each

WHAT DRIFTED	REAL CASE	THE TEST THAT SHOWS IT
the words	0.4871541501976284 retold as "≈ 0.49"; both models skipped irrigation §21	s ≠ v (only if the reader dereferences)
the reader	162 of 200 pre-fix S2 records carry the pixel south §7 §8	DN(b) ≠ COG[<i>row</i>], [<i>col</i>]]
the source	Bengaluru elevation 918.0 → 915.07 m; "as known on 15 Jun" still returns 918.0 §12 §13	fn_key: ≠ fn_keys: (open_meteo_copdem90m → copernicus_dem_30m_aws_pixel)
the date	a raster asked for 23 Sep was the 25 Sep scene, no warning (section 8); a cube member asked for 15 May is the 10 Jun scene §6	date(scene) ≠ asked date
the place	a token re-labelled with another cell §4 ; /v1/ask read "Keylong" as the town point 597 m away and answered NDVI 0.28 from that cell §30	cell(b) ≠ cell(token) → refused
the signer	915.07 m signed 7 times: 7 fact_cids for one value §13	k(f ₁) = k(f ₂), cid(f ₁) ≠ cid(f ₂)
the referent	"Maasvlakte ramp 7" resolves to all of Maasvlakte, ≈ 12 × 10 km (emem:entity:4ity1z3k...)	none: an entity token names a label, not bytes

Six of the seven are one equality test on a field of a recomputable record. For an embedding the only test is e₁ ≠ e₂; it shows the vector moved, not which input did.

9 · SINCE THE PAPER WAS ACCEPTED

emem retired its foundation-model encoders

TESSERA: e = f32(*q₁q₂*) × f32(scale), 128 dims, recomputable only as a re-read · device: trace_root = Merkle(segment digests); a fact binds = BLAKE3(CBOR(value)) ∈ outputs; admit → no reject reason of 17

The accepted paper builds on embeddings. Since then Clay v1.5, Prithvi-EO-2.0, Galileo and the JEPA-v2 head were removed from the code [§23](#), and TESSERA is frozen on emem.dev; vectors signed before still resolve and verify [§24](#).

The records show the difference. The TESSERA vector [§9](#) re-reads bit-exact from its public product (128 B int8 + 4 B scale, max |Δ| = 0, 4 of 4), but it is attester_only; nobody can recompute the encoder step. The Clay vector [§10](#) names its weights by BLAKE2b, and the code that ran them is gone. A raw band read [§4](#) is recomputable by anyone from public bytes.

A device that cannot be recomputed must bind its output inside a complete OS execution trace. That gate ships as code, with conformance vectors; it admits no real hardware yet [§26](#).

11 · EMEMIFY YOUR WORK

Five steps to sign and cite your own values, each run on 29–30 Sep

pointer.v1: leaf = BLAKE3(u1 || u64 offset || u64 length || h), node = BLAKE3(1 || 1 || 1), odd node promoted · derive accepted = ulp_gap(v_{caller}, v_{emem}) = 0 (≤ 4 for sum, mean, n > 2)

1 min

10 min

1 hour

1 day

1 week

Give your agent the MCP server: claude mcp add --transport http emem https://emem.dev/mcp (18 core tools, 114 in all).

Cite a value in a paper by its token and receipt; a Claude skill (research-grade-citation) drafts the methods text.

Register your own result, signed by your key: an NDVI change of −0.1298 (25 Sep 2025 → 2026) at this cell [§11](#); emem recomputed it: bit-identical.

Hash your files in place: a signed pointer note lists each chunk's BLAKE3. This board is one [§1](#).

Run your own note: docker run ghcr.io/vortx-ai/emem. A local node re-read this pixel: bit-identical NDVI, a different fact_cid (its own key signs).

Apache-2.0, pip install ememdev · npm i @vortxai/emem · agent-to-agent (A2A) card at emem.dev/well-known/agent-card.json, signed.

12 · WHAT IS NOT ESTABLISHED

What a signature does not show

draft check allows → 3 number a: round(a·10^d)/10^d = s, d = decimals written, s = signed value → "≈ 0.49" passes for 0.4871

A signature says who served a value, not that it is true.

A fact_cid names one signed record; two signers of the same value mint different ids.

Place and object (entity) tokens are labels; only fact, raster, cube and tree tokens hash bytes; a bundle hashes a citation list.

A signed "no tile here" [§15](#) is a claim; no complete tile index backs it.

emem's draft checker rejects "35 °C" citing a signed 28.0 °C [§14](#), but it judges at the writer's precision and allows "≈ 0.49"; it does not check band or unit.

Re-minting a raster yields a new token for identical pixels (signed_at is hashed).

One responder and one operator; the only domain-vouched log co-signer, geo.qa, is also Vortx AI (109 other witness keys are unvouched). No read federation, no hardware attestation, nothing in orbit.

The agent experiments are single-shot samples without independent replication.

THE TOKEN FAMILY: EVERY DATA NEED, ONE GRAMMAR, ONE HASH RULE			
DATA NEED	TOKEN	ITS ID IS	HERE
a place	emem:cell:<cell164>	the quantised lat/lng itself; a view of every fact held there	§29
one value	emem:fact:<cell>:<cid>	BLAKE3(CBOR(fact))	§4 §12
nothing there	emem:fact: · kind absence	same rule; reason_cid = BLAKE3(reason)[−16]	§15
an embedding	emem:fact: · 128 / 1024 floats	same rule; attester_only	§9 §10
a derived value	emem:fact: · op, parents	same rule; emem re-runs op(parents)	§11
a raster	emem:raster:<aoi>:<band>:<tslot>:<d>	aoi = BLAKE3(CBOR(bbox)); pixels = BLAKE3(grid); d = cid of the derivation	§5
a raster through time	emem:cube:<aoi>:<band>:<cte..ti>:<d>	BLAKE3(CBOR([member d, tslot order]))	§6
rasters for one purpose	emem:raster:<set>:<d>	BLAKE3(CBOR([d., "purpose:p"]))	no receipt yet
many facts, one handle	emem:bundle:<cid>	BLAKE3(fact_cids joined by "[") [−16]	§16
a file, one chunk	emem:tree:<file>#row=i	Merkle root over (u1, offset, length, BLAKE3(chunk))	§1–3 §17–26
a reasoning stage	emem:state:<cid>	BLAKE3(CBOR{kind, previous state, fact_cids, payload})	§30
a named object	emem:entity:<cid>	hash of an anchor (e.g. an OSM id): a label, not bytes	panel 7
a device run	emem:trace:<cid>	BLAKE3(CBOR(signed OS trace)); no hardware admitted yet	panel 9
a chain of evidence	track.v1 note	link _i = BLAKE3(link _{i−1} ref _i)[−16]	this board

An embedding is one row of fourteen. Only fact, raster, cube and tree ids hash the observed bytes; bundle, raster, state and track ids hash lists of other ids; an entity id hashes a name. Grammars and hash rules as coded at emem 18adb67 (research/repro/v10/algorithms.mnd).

1 · FROM SCENE TO TOKEN

1.17 MB read from a 2.02 GB scene; the agent receives 84 characters

pixel: col = [i + (E − X)/s_x], row = [j + (Y − N)/s_y] in UTM 43N · p = (DN + o)·10^{−4}, o = −1000 if baseline ≥ 04.00 (PC) · NDVI = (p_s − p_x)/(p_s + p_x) · SCL ∈ {0,1,8,9,10} → next scene

Sentinel-2 L2A scene, upstream (all assets)

bytes emem range-read for this value (3 COG heads + 3 tiles)

signed record (fact), CBOR

token handed to the next agent (characters)

2,023,818,762

1,165,033

1,115

84

10³

10²

10¹

10⁰

10^{−1}

10^{−2}

10^{−3}

10^{−4}

10^{−5}

10^{−6}

10^{−7}

10^{−8}

10^{−9}

10^{−10}

10^{−11}

10^{−12}

bytes (log scale)

By our reconstruction of its read path, emem range-reads **0.058 %** of the scene to sign one NDVI value; three 64 KiB COG headers and one tile each of B04, B08 and SCL. It stores the 1,115-byte record, not the tiles. The next agent receives 84 characters (46 GPT-4 tokenizer tokens).

We hashed the Earth Search (AWS) copies of the B08 and B04 COGs into two signed pointer notes [§2](#) [§3](#): all 292 chunks of each (header + 291 tiles; 238.7 MB, 232.9 MB) under one Merkle root; the files stay on AWS. The tile holding the cell is emem:tree:kh1qtqrdbb6jponqn4gv72if7e#row=108 (row = leaf index); its 1,586,895 bytes hash to that row. These are not the Planetary Computer file the NDVI was read from (281.9 MB, offset not pre-subtracted), and emem records no hash of that file.

```
$4 emem:fact:defi.zb572.xoso.zbl1ec:0j5ceccile62vmm6hedpuk67cjt2pqcz733mtsuyffgakxbmyaa
band indices.ndvi day 20721 since 1970-01-01 = 25 Sep 2026
scene S2A_MSIL2A_20260925T054251_R00S_T43SFS · scene class 4, vegetation
DN B08 3502 B04 1900 B0A offset −1000 (PC, baseline ≈ 04.00)
value 0.4708994708994709 (B08−B04)/(B08+B04) after offset: bit-identical
signer 777ex3yi.. signed_at 2026-09-28T09:06:56Z · 1,115 B · BLAKE3 → 0j5cecci..
```

Earth Search serves the same tile with the offset already subtracted (B08 2502, B04 900); NDVI is identical. Sizes: Planetary Computer asset sizes (HTTP HEAD) and range reads, 30 Sep; the pointer notes and the 1,586,895-byte tile are Earth Search copies.

4 · WHAT THE TRACE FOUND

Before 28 Sep, 162 of 200 sampled records carry a neighbouring pixel

pre-fix row = round(j + (Y − N)/s_y): the south pixel whenever the fraction ≥ ½ · fixed: [·] · prevalence 162/200, Wilson score interval, z = 1.96

23 Sep scene · kxjvfwpa... before the fix

25 Sep scene · 0j5cecci... after the fix

0.637 0.619 0.518 0.420 0.359

0.576 0.542 0.554 0.476 0.392

0.427 0.437 0.486 0.539 0.475

0.319 0.327 0.344 0.421 0.478

0.114 0.078 0.107 0.217 0.419

0.591 0.651 0.491 0.414 0.369

0.527 0.547 0.502 0.500 0.386

0.395 0.425 0.471 0.538 0.444

0.300 0.332 0.302 0.405 0.477

0.095 0.062 0.074 0.209 0.428

● query point blue: pixel containing it dashed red: pixel whose DNs were signed

Re-reading the COGs named inside the signed records showed that the 23 Sep record [§7](#) carries the DNs of the pixel **10 m south** (dashed red: 2993/1972, SCL 5 bare soil, NDVI **0.3444**). The pixel that contains the point (blue) reads 0.4860. emem had rounded the fractional pixel index; GDAL takes the floor. The fix shipped on 28 Sep [§8](#).

AUDIT OF PRE-FIX SENTINEL-2 RECORDS

random sample, one per cell, of pre-fix S2 records cited on emem.dev's agent channel: 200 cells, 164 scenes, 19 bands (192 Element84, 8 Planetary Computer)	200
DNs match the rounded (neighbour) pixel, not the floor pixel	162 · 81 % [75–86 %]
rounding and floor give the same pixel	38
DNs match the floor pixel where the two rules differ	0
ΔNDVI in emem's signed change record at this cell, 23 → 25 Sep	+0.127
ΔNDVI at the containing pixel	−0.015

Re-reading the pixel found the error; the signatures date it. No signed byte was rewritten: old tokens still resolve to exactly what was signed, so the error is reproducible and datable. Post-fix records (after 28 Sep) read the containing pixel. Wilson 95 % interval. Figure: ● query point, blue = pixel containing it, dashed red = pixel emem read. The offset is a reproducibility error, not a claim about geolocation accuracy.

6 · COMPACTION: WHY A PARAPHRASE IS NOT A CITATION

After a shared summary, 0 of 72 answers were right, yet 3 of 36 pairs agreed

H₁: P(pair agrees) > P(answer correct) · Fisher exact one-sided on 8/72 → p = 0.035 · rule: irrigate = NDVI < 0.488

rule: irrigate if NDVI < 0.488

token 0.4871541501976284 → both irrigate

prose "≈ 0.49" → both skip

0.480

0.485

0.490

0.495

PRE-REGISTERED, GEMMA-4-12B + QWEN2.5-7B, ONE HOST

full context (control)

shared summary, compaction under pressure

shared summary, no pressure

wrong-but-agreeing pairs (3/36) exceed correct answers (0/72): Fisher one-sided p = 0.035 [§17](#); handoff n = 20 per arm: prose 2, dense retrieval 8, BM25 20, token bundle 20 [§18](#)

The trap value 0.4871541501976284 [§21](#) is the value observed 17 Jul and signed 18 Jul. It is a pre-fix read of the pixel south (section 4); it tests fidelity to the cited value, not the ground. BM25 ties the bundle on short handoffs; no-pressure arm p = 0.109 (n.s.); six scoring bugs were corrected across two scorers. Related: Perez et al., ICLR 2025; arXiv 2608.11242; arXiv 2607.25066.

CORRECT

AGREE

72/72

0/72

20/72

36/36

3/36

15/36

8 · RAW BANDS, THE AGENT'S OWN METHOD

Asked for 23 Sep raw bands, emem's raster tool served the 25 Sep scene

scene(t) = newest S2 item with cloud < c in [t − D, min(t + D, now)], (c, D) ∈ (40 %, 30 d), (60 %, 60 d), (80 %, 90 d) · t = 23 Sep → window 24 Aug – 30 Sep → 25 Sep (10.8 %); the 23 Sep scene (20 %) cannot win

10 PRE-REGISTERED RUNS, CLAUDE SONNET 5.5, EMEM AS THE ONLY MCP SERVER

chose their own bands and formula: B04, B08, (B08−B04)/(B08+B04)

noticed that both dates had returned one scene

got 23 Sep values (via recall and backfill): ΔNDVI 0.486 → 0.471 = −0.015, "no material change" [§27](#) [§28](#)

wrote "no material change" as a placeholder, saying it was not measured

answered "undetermined", outside the allowed answers

said "greener", the answer the pre-fix record gives (+0.127)

wrongly said no 23 Sep scene exists

cited ids that re-hash · data values traced to tool results

30/30 · all

The failure was emem's: emem_band_raster returned the byte-identical 25 Sep raster with no warning. The scene named inside the record is how nine agents saw it. No run used SCL.

Pre-registration blake3 30d8a1a1... hashed 13:49Z, 30 Sep, before trial 1; write tools denied. The primary test (does the pre-fix record mislead?) could not run; emem's read tools signed new records mid-run, so trials 9–10 were not independent. research/repro/data/v9/rawband · [§22](#)

10 · WHERE EMEM LOST

The authors' own scorecard

exact = answer = signed value · confidently wrong = answered A answer ≠ signed value · BM25 and dense retrieval top-5 over the same corpora	
CLAIM, AS SCORED BY THE AUTHORS §19	VERDICT
survives compaction where a paraphrase does not	supported
beats plain context when the value fits	refuted
beats BM25 on these corpora (BM25 16/16)	refuted
single tokens save context (they cost 9.5× the LLM tokens)	refuted
model agreement is evidence of correctness	refuted, p = 0.035
a bundle saves context (38 chars for up to 256 facts)	supported
beats peer memory products	not tested

Dereferenced citations: **99.2 %** exact, 0 confidently wrong (84.4 % before four fixes); dense retrieval 4/142, up to 138 confidently wrong by a median 252 m. On a retrieval miss one model abstained 74/96 and the other gave a real neighbouring-cell value 93/96 [§25](#).

Scope: 5 sites, 2 open 7–12B models, one host, no independent replication (labelled SAMPLE). The record lists nineteen withdrawn claims [§20](#).

13 · SEAL

Check this board

link = cid(""), link_i = base32(BLAKE3(link_{i−1} || ref_i)[−16]); head = link_n



Opens in ememdemo, which re-hashes every 5-step in your browser and recomputes the track's hash chain.

track: emem.dev/memories/by_attester/njedkg1t/gfkucqcmhvsbse1cyoxjkwoe.md
head: sxmIncauzwifzemaahyo7mpktt (30 of 30 steps)
board.jpg sha256: 42690cb98dd00e7c1b1a2aa05e4ecba9aedf1dde67faef87f8bb77b18ddd c0f7
written after log size 2,573,960 (30 Sep 2026 22:49 UTC)
logged as entry 2,573,960

A file cannot contain its own hash, so this box is the only part of the board not inside the published bytes [§1](#). Signed by the poster key njedkg1...; re-check scripts: research/repro/v8; each formula with its line in emem's code: research/repro/v10/algorithms.mnd.

5-steps: 1 board · 2–3 COG pointers · 4 NDVI record · 5 B04 raster · 6 cube · 7 pre-fix record · 8 CHANGELOG line 68 · 9 TESSERA · 10 Clay · 11 your derivation · 12–13 elevation · 14 temperature (draft check) · 15 absence · 16 bundle of 8 · 17–20 doc rows at emem 213e273 · 21 trap value · 22 our measurements · 23–26 doc rows at emem 18adb67 · 27–28 23 Sep B04, B08 (post-fix) · 29 the Keylong cell · 30 an /v1/ask reasoning stage. Traffic to 30 Sep: 96,728 MCP tool calls; PyPI 314 · npm 304 last month; 96.2 % of signed agent notes are from operator keys

Vortx AI